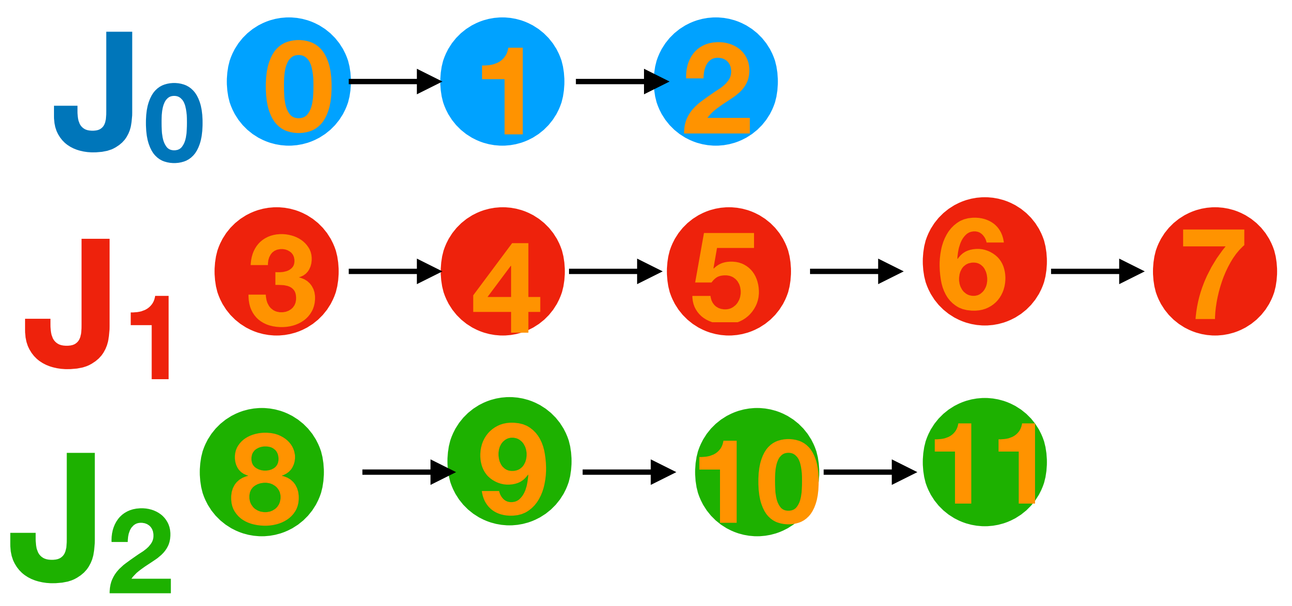
Metaheuristic Approach



ENCODAGE (construction de la solution gène)

...

ENCODAGE (construction de la solution par chromosome)

1. Tris topologique de toutes les opérations possibles (stochastique)
2. Affectation des premiers workers aléatoirement (niveau requis)
3. Affectation des second workers aléatoirement (au moins diff de 1 si pas le niveau)

operation sequence chromosome (OSC)	3	4	0	8	9	1	2	5	6	7	10	11
first worker assignement chromosome (FWAC)	0	2	2	-1	0	0	1	1	0	2	0	2
second worker assignement chromosome (SWAC)	2	-1	0	-2	2	1	-1	-1	-1	1	2	-1

W_0 W_1 W_2

Mode of operation **deducted** with level of worker Need for calculate new skills level and cognitive load	C	A	T	NA	C	C	AL	A	A	T	T	A

A : Alone
C : Collab
T : Teaching
AL : Alone no Level
NA : Not Assigned

DECODAGE

1. Définir vecteur du mode opératoire des opérations
2. Matrice (nb_w x nb_task) pour définir date de début/fin des opérations
    1. Savoir job/task fait avant et après une limite de temps

VOISINAGE

1. En gardant ordre des opérations
    1. Choix aléatoire d'un gène et voir les duos de workers qui améliore la fitness
2. Intervertir 2 opérations puis réaffecter duos de workers de toutes les opérations (ou seulement opération intervertit)

Metaheuristic Approach

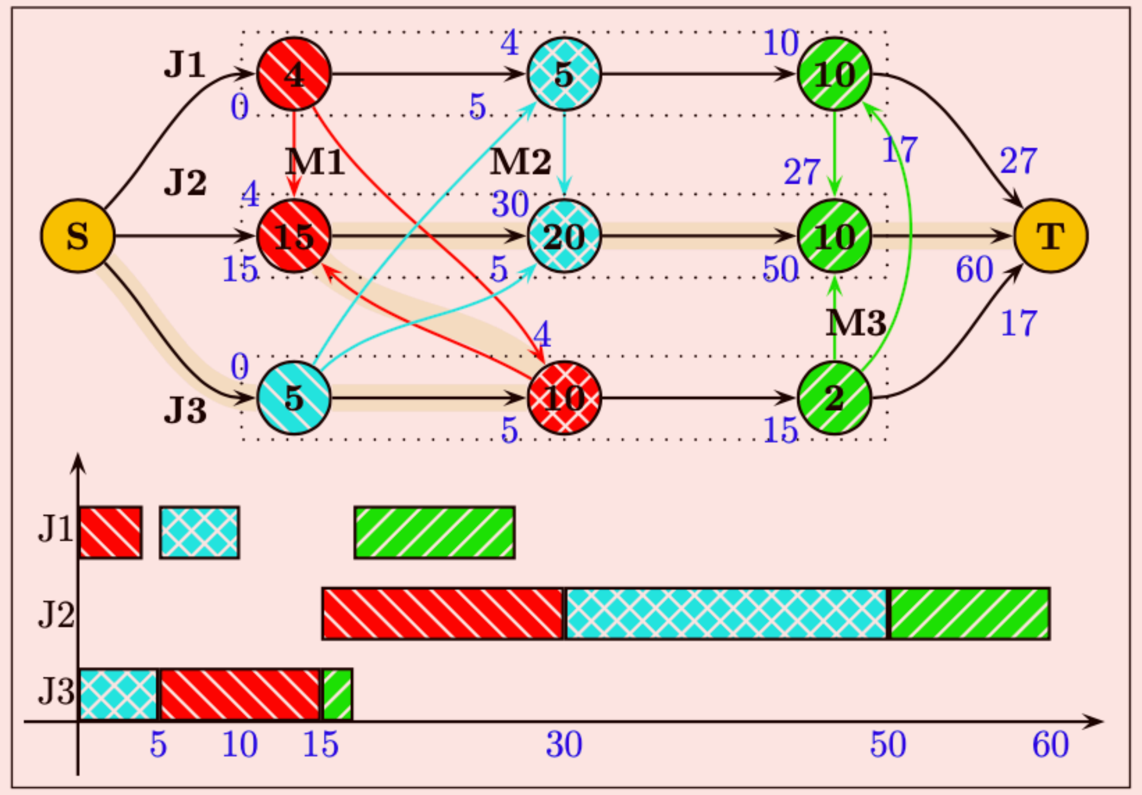


FIGURE 3. Disjunctive graph: critical path with a makespan of 60.

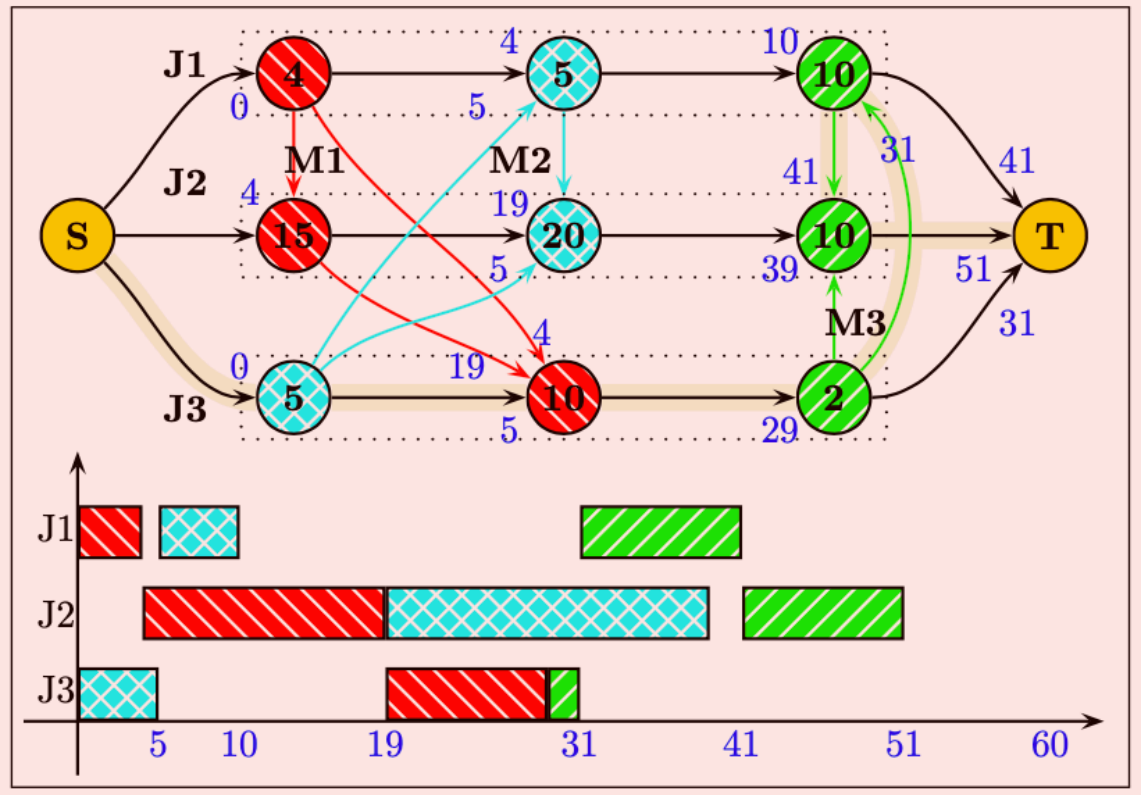


FIGURE 4. Disjunctive graph: recomputing the critical path with a makespan of 51.

S. Binato, W.J. Hery, D. Loewenstern, and M.G.C. Resende. A greedy randomized adaptive search procedure for job shop scheduling. In P. Hansen and C.C. Ribeiro, editors, Essays and surveys on metaheuristics. Kluwer Academic Publishers, 2001

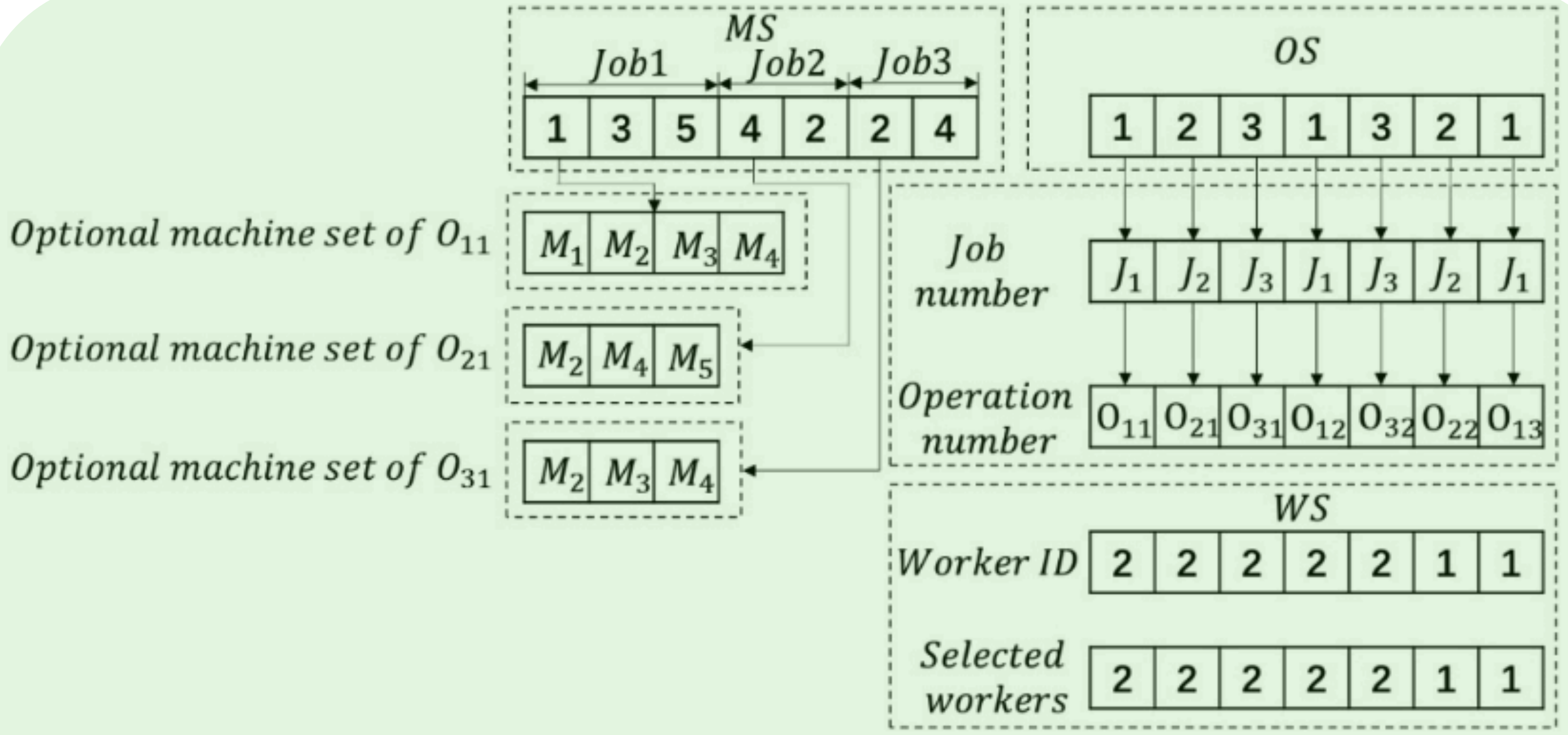


Fig. 4. Code schematic.

Wang, Chuang, Di Fan, Yang Liu, Shan Ren, et Jin Wang. « Dual-resource flexible job shop scheduling considering worker proficiency differences ». *Computers & Operations Research* 184 (décembre 2025): 107216. https://doi.org/10.1016/j.cor.2025.107216.

Wu, R., Li, Y., Guo, S. and Xu, W., 2018. Solving the dual-resource constrained flexible job shop scheduling problem with learning effect by a hybrid genetic algorithm. *Advances in Mechanical Engineering*, *10*(10), p.1687814018804096.

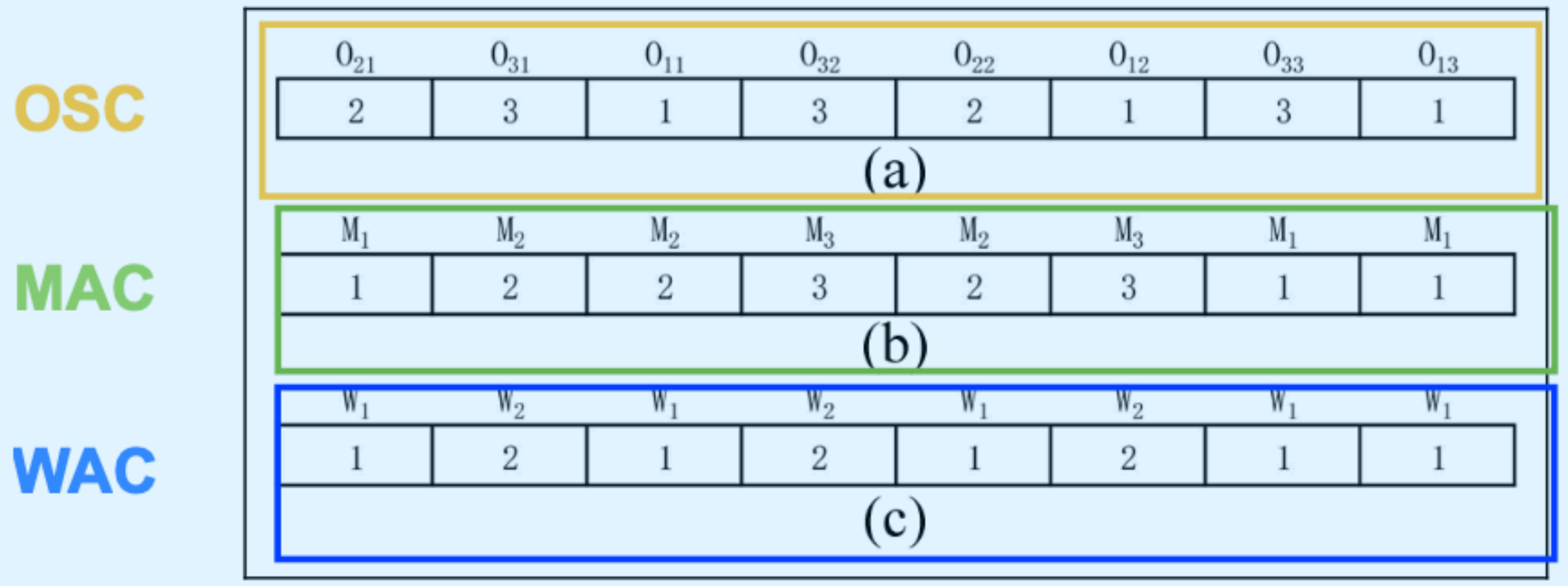


Figure 1. Individual coding example: (a) operation sequence chromosome, (b) machine assignment chromosome, and (c) worker assignment chromosome.

Table 3.1: Decoding Methods - quadruple string

(1,1,4,2)	(2,1,2,2)	(4,1,1,1)	(4,2,5,1)	(2,2,1,2)	(3,1,2,2)	(1,2,3,1)	(1,3,1,1)	(3,2,1,2)	(3,3,2,2)

Table 3.2: Single Chromosome encoding

1	2	4	4	2	3	1	1	3	3

Table 3.3: Two Vector encoding

OSV:	$O_{1,1}$	$O_{2,1}$	$O_{4,1}$	$O_{4,2}$	$O_{2,2}$	$O_{3,1}$	$O_{1,2}$	$O_{1,3}$	$O_{3,2}$	$O_{3,3}$
RAV:	(M_4,W_2)	(M_2,W_2)	(M_1,W_1)	(M_5,W_1)	(M_1,W_2)	(M_2,W_2)	(M_3,W_1)	(M_1,W_1)	(M_1,W_2)	(M_2,W_2)

Table 3.4: Three Layered Chromosome

OSV:	1	2	4	4	2	3	1	1	3	3
MAS:	4	2	1	5	1	2	3	1	1	2
WAS:	2	2	1	1	2	2	1	1	2	2

Marta, D.M.A., 2021. Dual Resource Constrained Flexible Job Shop Problem using Hybrid Genetic Algorithm.